

Solar N appendix

Sun · N · downloadable appendix

Solar nitrogen: 7.883 ± 0.182

Asplund, Amarsi & Grevesse 2021: 7.83 ± 0.07 · this measurement +0.053 dex

The solar N I lines are weak (0.3–1% deep) and carry CN absorption inside their own profiles, so the result depends on molecular opacity and on continuum placement at the 0.1% level. Both are now modelled: CN is synthesised with the line, and the continuum is placed locally around every line. For comparison, Asplund et al. 2021 give 7.77 from N I after empirically rescaling the CN blends, and Lodders, Bergemann & Palme 2025 adopt 7.94 ± 0.11 from the two least-blended lines. The error bar carries the full uncertainty budget, including CN blending and telluric residuals.

IAG FTS · NIR · Reference Grade · CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE · $n = 244$. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

red-optical

7.904 ± 0.188

± 0.010 stat · ± 0.188 syst

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

NIR

7.883 ± 0.182

± 0.040 stat · ± 0.177 syst

CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE · IAG FTS · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	12 products -- CN A–X (0–0) band, near-infrared (molecular band); N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025)
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A–C, or the paper's own stated error), between 5% and 60% deep	1 product -- CN red system, visible (molecular band; diagnostic only)
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS

A(N) dex; axis 3.93-10.66

CN red system, visible (molecular band; diagnostic only) · 1D LTE
Codex Grade · 194 lines



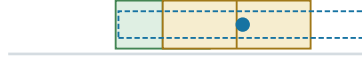
7.296
±3.338 stat ±1.567 syst

VIS axis: 3.93 · 5.61 · 7.30 · 8.98 · 10.66 A(N) dex

red-optical — IAG FTS

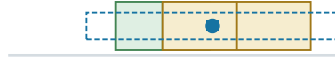
A(N) dex; axis 7.60-8.24

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D LTE
Reference Grade · 2 lines



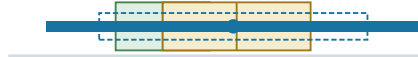
7.949
±0.002 stat ±0.185 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
3D NLTE (Amarsi)
Reference Grade · 2 lines



7.904
±0.010 stat ±0.188 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D NLTE (Bergemann)
Reference Grade · 1 line

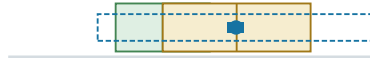


7.935
±0.278 stat ±0.200 syst

red-optical — Kitt Peak (Kurucz 2005)

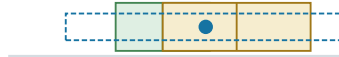
A(N) dex; axis 7.60-8.24

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D LTE
Reference Grade · 2 lines



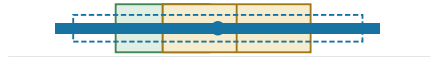
7.939
±0.013 stat ±0.206 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
3D NLTE (Amarsi)
Reference Grade · 2 lines



7.894
±0.005 stat ±0.208 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D NLTE (Bergemann)
Reference Grade · 1 line

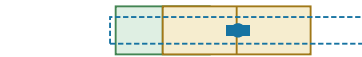


7.912
±0.241 stat ±0.215 syst

red-optical — Kitt Peak (1984, molecfit)

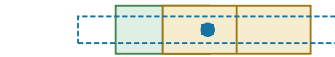
A(N) dex; axis 7.60-8.24

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D LTE
Reference Grade · 2 lines



7.942
±0.018 stat ±0.190 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
3D NLTE (Amarsi)
Reference Grade · 2 lines



7.897
±0.010 stat ±0.193 syst

N I 8629 + 8683 Å, the two least-blended lines
(Magg et al. 2022, adopted by Lodders et al. 2025) ·
1D NLTE (Bergemann)
Reference Grade · 1 line



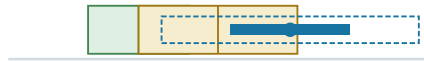
7.910
±0.280 stat ±0.235 syst

red-optical axis: 7.60 · 7.76 · 7.92 · 8.08 · 8.24 A(N) dex

NIR — CRIRES+ J

A(N) dex; axis 7.65-8.25

CN A-X (0-0) band, near-infrared (molecular band) ·
1D LTE
Reference Grade · 185 lines



8.040
±0.083 stat ±0.178 syst

NIR — IAG FTS

A(N) dex; axis 7.65-8.25

CN A-X (0-0) band, near-infrared (molecular band) ·
1D LTE
Reference Grade · 244 lines

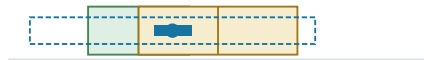


7.883
±0.040 stat ±0.177 syst

NIR — Kitt Peak (1984, molecfit)

A(N) dex; axis 7.65-8.25

CN A-X (0-0) band, near-infrared (molecular band) ·
1D LTE
Reference Grade · 215 lines



7.877
±0.026 stat ±0.198 syst

NIR axis: 7.65 · 7.80 · 7.95 · 8.10 · 8.25 A(N) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every nitrogen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the nitrogen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C₂, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(N)	Largest uncertainty terms (dex)
VIS	CN red system, visible (molecular band; diagnostic only)	HARPS	1D-LTE	194	7.296 (1D-LTE)	fit / line-to-line scatter 3.34, continuum placement 1.34
red-optical	N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025)	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1-2	7.894-7.904 (3D-NLTE)	continuum placement 0.18, microturbulence 0.03

Band	What was measured	Solar spectra	Treatments	Lines	Result A(N)	Largest uncertainty terms (dex)
NIR	CN A-X (0-0) band, near-infrared (molecular band)	CRIRES+ J, IAG FTS, Kitt Peak (1984, molecfit)	1D-LTE	185-244	7.877-8.040 (1D-LTE)	line strengths (gf) 0.12, spread between spectra 0.09

Solar nitrogen context

The nitrogen strategy uses red-optical N I as the primary atomic route, with NH providing an independent molecular check and CN a carbon-dependent cross-check. An unresolved or unconstrained fit supplies no adopted Solar nitrogen abundance.

[C/O indicator strategy](#) · [Methodology](#) · [Nitrogen strategy](#)

Open provenance decisions

NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation.

[Adjudication record](#) · [Indicator gf checks](#)

Download the evidence

[Download PDF — Solar N appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned science snapshot. Source commit c730d5e72e3f8902d1d4d800b46fe89fbcafee69
Feed N.json v1.72 · updated 2026-10-03T22:11:26Z
Generated 2026-10-09T16:35:54-06:00 (source commit timestamp)
Feed SHA-256 74919efb0dc55af06bd2059bfbe9e51e769cd8e033b2a5f1624962ea807a60da
Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar N atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Atomic data and gf authorities. The recorded N I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [\[3\]](#), [\[4\]](#).

Atomic data and gf authorities. NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation. Sources: [\[3\]](#), [\[5\]](#).

Model and grid sources. The N I registry uses the Amarsi GALAH/PySME 1D-NLTE departure grid. The solar 3D-NLTE atomic analysis is a separate source, not a full-3D correction applied by this feed. Sources: [6], [7], [8].

Molecular line data and delivery. CN line-data delivery: CN red A-X: Brooke et al. 2014 / Sneden et al. 2014 (isotopologues). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpgdl.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [9], [10], [11].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Atomic data and gf authorities. Atomic N I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed label: Synth · 3D-NLTE · Amarsi. Cited per-line solar correction, Amarsi et al. 2020 Table 3 (3D non-LTE minus 1D LTE); not an in-transfer 3D-NLTE result. Run provenance: Amarsi, Grevesse, Grumer, Asplund, Barklem & Collet 2020, A&A 636, A120, Table 3 (solar, disk-centre EWs) -- 3D non-LTE (STAGGER) - 1D LTE (MARCS); SOLAR per-line additive correction, same EWs in both Sources: [6], [7], [8].

Model and grid sources. Feed label: Synth · 1D-NLTE · Bergemann. Registered source: Amarsi/PySME N I 1D-NLTE. The generic Bergemann label is retained as supplied and is not used as a bibliographic attribution. Sources: [6], [7], [8].

Product-to-source index

NIR · CRIRES+ J · CN A-X (0-0) band, near-infrared (molecular band) · 1D LTE · [1], [2], [9], [10], [11]
N-products-f80b21113bb7

NIR · IAG FTS · CN A-X (0-0) band, near-infrared (molecular band) · 1D LTE · [1], [2], [9], [10], [11]
N-products-5be1014384a4

NIR · Kitt Peak (1984, molecfit) · CN A-X (0-0) band, near-infrared (molecular band) · 1D LTE · [1], [2], [9], [10], [11]
N-products-dc37e3be96d5

VIS · HARPS · CN red system, visible (molecular band; diagnostic only) · 1D LTE · [1], [2], [9], [10], [11]
N-products-7886ff44426f

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-ba33eb2559

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-34b552f38f15

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-products-dacd11127c03

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-b8f600945b9a

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-5f0e1db99ddf

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8]
N-products-0dfb1baa2bbc

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-af27659f2bea

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-74c0cd7f098d

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8]

N-products-e11f215d2461

red-optical · IAG FTS · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#), [\[7\]](#), [\[8\]](#)
N-quarantine-fc1525ccd677

red-optical · Kitt Peak (Kurucz 2005) · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#),
[\[6\]](#), [\[7\]](#), [\[8\]](#) N-quarantine-9cb72bc628d8

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)
N-quarantine-35945ad2b73d

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#),
[\[7\]](#), [\[8\]](#) N-quarantine-4d3667606a8f

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#),
[\[6\]](#), [\[7\]](#), [\[8\]](#) N-quarantine-66d14a326159

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Tachiev & Froese Fischer 2002, *A&A* 385, 716 DOI: 10.1051/0004-6361:20011816. [source record](#)

Model and grid sources

[6] — Amarsi et al. 2020, *A&A* 642 A62 DOI: 10.1051/0004-6361/202038650. [source record](#)

[7] — Amarsi, Grevesse, Gruber, Asplund, Barklem & Collet 2020, *A&A* 636, A120 DOI: 10.1051/0004-6361/202037890. [source record](#)

[8] — Amarsi 'Grid/NLTE' Zenodo, version 3 (the original 13-element 2020 release) DOI: 10.5281/zenodo.3982506. [source record](#)

Molecular line data and delivery

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. *A&A* 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

[10] — Brooke et al. 2014, *ApJS* 210, 23 DOI: 10.1088/0067-0049/210/2/23. [source record](#)

[11] — Sneden, C.; Lucatello, S.; Ram, R. S.; Brooke, J. S. A.; Bernath, P. (2014). Line Lists for the Red and Violet Systems of CN, ¹³C¹⁴N, and ¹²C¹⁵N, and Application to Astronomical Spectra. *ApJS* 214, 26. DOI: 10.1088/0067-0049/214/2/26. [source record](#)